

SPENVIS 4.6.14.358210-May-2026 01:19:45

Solar cell damage equivalent fluences

Project: GEO_10Y

geo_10yers_duration

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Cell: Si
Electron/proton damage ratios: P _{max} = 3000.0 V _{oc} = 3000.0 I _{sc} = 3000.0
Coverglass
material: fused silica density: 2.20 (g/cm ³)

Summary of 1 MeV equivalent electron fluences (cm⁻²)

Coverglass thickness			Total			Trapped electrons	Trapped protons			Solar protons		
g cm ⁻²	mils	micron	P _{max}	V _{oc}	I _{sc}	P _{max} , V _{oc} , I _{sc}	P _{max}	V _{oc}	I _{sc}	P _{max}	V _{oc}	I _{sc}
0.0000	0.0	0.00	1.543E+18	1.543E+18	2.815E+16	2.800E+14	1.527E+18	1.527E+18	2.526E+16	1.487E+16	1.487E+16	2.605E+15
0.0056	1.0	25.40	4.738E+15	4.738E+15	1.791E+15	2.394E+14	1.748E+15	1.748E+15	4.137E+14	2.750E+15	2.750E+15	1.138E+15
0.0168	3.0	76.20	1.649E+15	1.649E+15	8.694E+14	1.955E+14	7.620E+13	7.620E+13	2.537E+13	1.377E+15	1.377E+15	6.486E+14
0.0335	6.0	152.40	9.603E+14	9.603E+14	5.726E+14	1.549E+14	7.511E+11	7.511E+11	2.686E+11	8.046E+14	8.046E+14	4.174E+14
0.0671	12.0	304.80	5.427E+14	5.427E+14	3.608E+14	1.078E+14	1.196E+09	1.196E+09	4.033E+08	4.349E+14	4.349E+14	2.530E+14
0.1118	20.0	508.00	3.257E+14	3.257E+14	2.365E+14	7.351E+13	1.000E-12	1.000E-12	1.000E-12	2.522E+14	2.522E+14	1.630E+14
0.1676	30.0	762.00	2.187E+14	2.187E+14	1.681E+14	4.938E+13	1.000E-12	1.000E-12	1.000E-12	1.694E+14	1.694E+14	1.187E+14
0.3353	60.0	1524.00	1.014E+14	1.014E+14	8.295E+13	1.882E+13	1.000E-12	1.000E-12	1.000E-12	8.259E+13	8.259E+13	6.413E+13
User defined coverglass thickness												
0.0220	3.9	100.00	1.334E+15	1.334E+15	7.381E+14	1.784E+14	1.245E+13	1.245E+13	4.264E+12	1.115E+15	1.115E+15	5.456E+14

[Trapped protons](#) for [mission segment 1](#) for P_{max} (integration restricted to the input energy range). D(E,...) values are coarse averages introduced into the table to simplify the output information; the accuracy of the results should not be affected by this simplification.

			0.00 micron		25.40 micron		76.20 micron		152.40 micron		304.80 micron		508.00 micron		762.00 micron		1524.00 micron	
Energy Int. (MeV)	Int. Fluence (cm ⁻²)	Diff. Fluence (cm ⁻²)	D(E, 0.0)	Eq. Fluence (cm ⁻²)	D(E, 1.0)	Eq. Fluence (cm ⁻²)	D(E, 3.0)	Eq. Fluence (cm ⁻²)	D(E, 6.0)	Eq. Fluence (cm ⁻²)	D(E, 12.0)	Eq. Fluence (cm ⁻²)	D(E, 20.0)	Eq. Fluence (cm ⁻²)	D(E, 30.0)	Eq. Fluence (cm ⁻²)	D(E, 60.0)	Eq. Flu (cm
0.10	0.20	7.209E+14	4.643E+14	6.11E-01	2.837E+14	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.20	0.30	2.566E+14	1.530E+14	7.78E-01	1.190E+14	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.30	0.40	1.036E+14	6.938E+13	9.19E-01	6.375E+13	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.40	0.60	3.421E+13	2.454E+13	1.11E+00	2.729E+13	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.60	0.80	9.673E+12	5.631E+12	1.40E+00	7.858E+12	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.80	1.00	4.042E+12	2.126E+12	1.66E+00	3.519E+12	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
1.00	1.20	1.916E+12	7.777E+11	1.88E+00	1.466E+12	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
1.20	1.30	1.139E+12	2.326E+11	2.04E+00	4.737E+11	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
1.30	1.40	9.061E+11	1.728E+11	2.12E+00	3.663E+11	9.63E-02	1.664E+10	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

[Trapped protons](#) for [mission segment 1](#) for V_{oc} (integration restricted to the input energy range). $D(E,...)$ values are coarse averages introduced into the table to simplify the output information; the accuracy of the results should not be affected by this simplification.

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[Trapped protons](#) for [mission segment 1](#) for I_{sc} (integration restricted to the input energy range). D(E,...) values are coarse averages introduced into the table to simplify the output information; the accuracy of the results should not be affected by this simplification.

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Trapped electrons for **mission segment 1** (integration restricted to the input energy range). D(E,...) values are coarse averages introduced into the table to simplify the output information; the accuracy of the results should not be affected by this simplification.

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0.17	0.18	3.348E+15	2.381E+14	1.19E-03	2.829E+11	2.30E-04	5.466E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
0.18	0.19	3.109E+15	2.211E+14	1.94E-03	4.294E+11	4.39E-04	9.703E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
0.19	0.20	2.888E+15	2.054E+14	2.98E-03	6.117E+11	7.95E-04	1.633E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
0.20	0.22	2.683E+15	2.879E+14	5.02E-03	1.445E+12	1.67E-03	4.794E+11	2.36E-04	6.793E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
0.22	0.24	2.395E+15	2.570E+14	8.40E-03	2.159E+12	3.58E-03	9.191E+11	7.68E-04	1.974E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
0.24	0.26	2.138E+15	2.294E+14	1.23E-02	2.818E+12	6.31E-03	1.447E+12	1.92E-03	4.404E+11	1.88E-04	4.306E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
0.26	0.28	1.909E+15	2.048E+14	1.71E-02	3.505E+12	9.72E-03	1.990E+12	3.81E-03	7.793E+11	6.44E-04	1.318E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
0.28	0.30	1.704E+15	1.828E+14	2.35E-02	4.287E+12	1.40E-02	2.566E+12	6.37E-03	1.164E+12	1.63E-03	2.987E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
0.30	0.32	1.521E+15	1.571E+14	3.04E-02	4.775E+12	1.94E-02	3.048E+12	9.61E-03	1.510E+12	3.21E-03	5.041E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
0.32	0.36	1.364E+15	2.672E+14	4.13E-02	1.102E+13	2.85E-02	7.611E+12	1.59E-02	4.238E+12	6.56E-03	1.753E+12	5.02E-04	1.342E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
0.36	0.40	1.097E+15	2.148E+14	5.93E-02	1.273E+13	4.35E-02	9.338E+12	2.72E-02	5.838E+12	1.36E-02	2.929E+12	2.47E-03	5.297E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
0.40	0.45	8.820E+14	1.923E+14	8.16E-02	1.569E+13	6.31E-02	1.214E+13	4.29E-02	8.245E+12	2.47E-02	4.753E+12	7.04E-03	1.355E+12	4.14E-04	7.951E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00
0.45	0.50	6.897E+14	1.504E+14	1.09E-01	1.639E+13	8.80E-02	1.323E+13	6.38E-02	9.587E+12	4.07E-02	6.118E+12	1.55E-02	2.335E+12	2.61E-03	3.919E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00
0.50	0.60	5.393E+14	1.726E+14	1.58E-01	2.732E+13	1.32E-01	2.282E+13	1.01E-01	1.749E+13	7.08E-02	1.223E+13	3.44E-02	5.932E+12	1.07E-02	1.852E+12	8.61E-04	1.487E+11	0.00E+00	0.000E+00
0.60	0.70	3.667E+14	1.178E+14	2.32E-01	2.734E+13	2.02E-01	2.379E+13	1.64E-01	1.931E+13	1.24E-01	1.465E+13	7.25E-02	8.537E+12	3.30E-02	3.892E+12	9.20E-03	1.084E+12	0.00E+00	0.000E+00
0.70	0.80	2.489E+14	6.594E+13	3.07E-01	2.026E+13	2.75E-01	1.817E+13	2.33E-01	1.538E+13	1.87E-01	1.234E+13	1.22E-01	8.065E+12	6.74E-02	4.446E+12	2.81E-02	1.853E+12	0.00E+00	0.000E+00
0.80	0.90	1.829E+14	4.318E+13	3.84E-01	1.659E+13	3.51E-01	1.516E+13	3.06E-01	1.319E+13	2.54E-01	1.098E+13	1.80E-01	7.773E+12	1.12E-01	4.849E+12	5.77E-02	2.491E+12	1.41E-03	6.101E-03
0.90	1.00	1.398E+14	3.299E+13	4.59E-01	1.515E+13	4.26E-01	1.404E+13	3.78E-01	1.248E+13	3.24E-01	1.068E+13	2.42E-01	7.976E+12	1.64E-01	5.400E+12	9.64E-02	3.180E+12	8.66E-03	2.856E-03
1.00	1.20	1.068E+14	3.882E+13	5.77E-01	2.242E+13	5.41E-01	2.099E+13	4.88E-01	1.894E+13	4.27E-01	1.657E+13	3.34E-01	1.298E+13	2.44E-01	9.466E+12	1.62E-01	6.271E+12	3.15E-02	1.223E-02
1.20	1.40	6.795E+13	2.539E+13	7.57E-01	1.923E+13	7.16E-01	1.817E+13	6.55E-01	1.663E+13	5.84E-01	1.484E+13	4.76E-01	1.209E+13	3.68E-01	9.352E+12	2.68E-01	6.801E+12	8.77E-02	2.227E-02
1.40	1.60	4.256E+13	1.600E+13	9.52E-01	1.523E+13	9.06E-01	1.449E+13	8.38E-01	1.341E+13	7.58E-01	1.213E+13	6.35E-01	1.016E+13	5.09E-01	8.151E+12	3.89E-01	6.231E+12	1.64E-01	2.629E-02
1.60	1.80	2.656E+13	1.000E+13	1.15E+00	1.153E+13	1.10E+00	1.104E+13	1.03E+00	1.031E+13	9.43E-01	9.434E+12	8.07E-01	8.068E+12	6.65E-01	6.650E+12	5.26E-01	5.262E+12	2.55E-01	2.555E-02
1.80	2.00	1.655E+13	6.318E+12	1.36E+00	8.574E+12	1.31E+00	8.251E+12	1.23E+00	7.768E+12	1.14E+00	7.175E+12	9.88E-01	6.239E+12	8.31E-01	5.250E+12	6.75E-01	4.267E+12	3.61E-01	2.279E-02
2.00	2.25	1.024E+13	4.430E+12	1.59E+00	7.037E+12	1.54E+00	6.804E+12	1.45E+00	6.443E+12	1.35E+00	6.000E+12	1.19E+00	5.292E+12	1.02E+00	4.537E+12	8.52E-01	3.776E+12	4.93E-01	2.182E-02
2.25	2.50	5.806E+12	2.564E+12	1.85E+00	4.750E+12	1.80E+00	4.610E+12	1.71E+00	4.388E+12	1.61E+00	4.115E+12	1.43E+00	3.675E+12	1.25E+00	3.202E+12	1.06E+00	2.719E+12	6.55E-01	1.680E-02
2.50	2.75	3.242E+12	1.321E+12	2.12E+00	2.795E+12	2.06E+00	2.721E+12	1.97E+00	2.603E+12	1.86E+00	2.457E+12	1.68E+00	2.218E+12	1.48E+00	1.958E+12	1.28E+00	1.689E+12	8.32E-01	1.099E-02
2.75	3.00	1.921E+12	7.618E+11	2.37E+00	1.807E+12	2.32E+00	1.764E+12	2.23E+00	1.696E+12	2.11E+00	1.610E+12	1.93E+00	1.467E+12	1.72E+00	1.310E+12	1.50E+00	1.146E+12	1.02E+00	7.769E-03
3.00	3.25	1.160E+12	4.175E+11	2.62E+00	1.096E+12	2.57E+00	1.072E+12	2.48E+00	1.034E+12	2.36E+00	9.859E+11	2.17E+00	9.053E+11	1.95E+00	8.162E+11	1.73E+00	7.225E+11	1.22E+00	5.074E-03
3.25	3.50	7.421E+11	2.687E+11	2.87E+00	7.708E+11	2.81E+00	7.557E+11	2.72E+00	7.308E+11	2.60E+00	6.993E+11	2.41E+00	6.462E+11	2.19E+00	5.872E+11	1.95E+00	5.247E+11	1.41E+00	3.795E-03
3.50	3.75	4.735E+11	1.535E+11	3.12E+00	4.786E+11	3.06E+00	4.698E+11	2.97E+00	4.554E+11	2.85E+00	4.369E+11	2.64E+00	4.058E+11	2.42E+00	3.712E+11	2.18E+00	3.343E+11	1.62E+00	2.479E-03
3.75	4.00	3.200E+11	9.667E+10	3.37E+00	3.256E+11	3.31E+00	3.200E+11	3.21E+00	3.107E+11	3.09E+00	2.988E+11	2.88E+00	2.788E+11	2.65E+00	2.563E+11	2.40E+00	2.323E+11	1.82E+00	1.759E-03
4.00	4.50	2.233E+11	1.015E+11	3.70E+00	3.758E+11	3.64E+00	3.699E+11	3.55E+00	3.601E+11	3.42E+00	3.475E+11	3.21E+00	3.262E+11	2.97E+00	3.019E+11	2.72E+00	2.759E+11	2.11E+00	2.138E-03
4.50	5.00	1.218E+11	4.554E+10	4.15E+00	1.892E+11	4.10E+00	1.866E+11	4.00E+00	1.822E+11	3.88E+00	1.765E+11	3.66E+00	1.668E+11	3.42E+00	1.557E+11	3.15E+00	1.437E+11	2.51E+00	1.145E-03
5.00	5.50	7.624E+10	2.340E+10	4.61E+00	1.080E+11	4.56E+00	1.066E+11	4.46E+00	1.044E+11	4.33E+00	1.014E+11	4.11E+00	9.627E+10	3.86E+00	9.043E+10	3.59E+00	8.405E+10	2.93E+00	6.851E-03
5.50	6.00	5.283E+10	1.450E+10	5.06E+00	7.342E+10	5.01E+00	7.260E+10	4.91E+00	7.116E+10	4.78E+00	6.928E+10	4.56E+00	6.605E+10	4.30E+00	6.234E+10	4.02E+00	5.828E+10	3.33E+00	4.831E-03
6.00	7.00	3.834E+10	3.834E+10	2.59E+00	9.915E+10	2.56E+00	9.815E+10	2.51E+00	9.640E+10	2.45E+00	9.409E+10	2.35E+00	9.011E+10	2.23E+00	8.551E+10	2.10E+00	8.045E+10	1.77E+00	6.794E-03
7.00	8.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
8.00	9.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
9.00	10.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
10.00	15.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
15.00	20.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
20.00	25.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
25.00	30.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
30.00	40.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
Equiv. 1 MeV Electron Fluence (cm ⁻²)					2.800E+14		2.394E+14		1.955E+14		1.549E+14		1.078E+14		7.351E+13		4.938E+13		1.882E-03

Solar protons for P_{max} (integration restricted to the input energy range). D(E,...) values are coarse averages introduced into the table to simplify the output information; the accuracy of the results should not be affected by this simplification.

			0.00 micron		25.40 micron		76.20 micron		152.40 micron		304.80 micron	
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3.40	3.60	3.798E+11	1.739E+10	2.48E+00	4.306E+10	2.42E+00	4.206E+10	1.34E+00	2.336E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
3.60	3.80	3.624E+11	1.590E+10	2.36E+00	3.758E+10	2.40E+00	3.822E+10	1.56E+00	2.485E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
3.80	4.00	3.465E+11	1.444E+10	2.23E+00	3.220E+10	2.35E+00	3.399E+10	1.74E+00	2.511E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.00	4.20	3.321E+11	1.316E+10	2.09E+00	2.751E+10	2.27E+00	2.986E+10	1.87E+00	2.466E+10	1.27E-01	1.677E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.20	4.40	3.189E+11	1.206E+10	1.96E+00	2.360E+10	2.16E+00	2.600E+10	1.96E+00	2.369E+10	4.09E-01	4.937E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.40	4.60	3.069E+11	1.121E+10	1.83E+00	2.049E+10	2.03E+00	2.272E+10	2.01E+00	2.250E+10	6.94E-01	7.781E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.60	4.80	2.957E+11	1.045E+10	1.71E+00	1.785E+10	1.90E+00	1.986E+10	2.00E+00	2.094E+10	9.50E-01	9.929E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.80	5.20	2.852E+11	1.916E+10	1.55E+00	2.962E+10	1.72E+00	3.304E+10	1.91E+00	3.660E+10	1.24E+00	2.383E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
5.20	5.60	2.661E+11	1.721E+10	1.36E+00	2.342E+10	1.52E+00	2.616E+10	1.74E+00	2.987E+10	1.51E+00	2.607E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
5.60	6.00	2.488E+11	1.548E+10	1.21E+00	1.867E+10	1.35E+00	2.086E+10	1.55E+00	2.405E+10	1.59E+00	2.468E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
6.00	6.40	2.334E+11	1.368E+10	1.08E+00	1.474E+10	1.20E+00	1.640E+10	1.39E+00	1.900E+10	1.53E+00	2.087E+10	1.55E-01	2.122E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
6.40	6.80	2.197E+11	1.232E+10	9.71E-01	1.197E+10	1.07E+00	1.323E+10	1.24E+00	1.533E+10	1.40E+00	1.728E+10	6.24E-01	7.689E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
6.80	7.20	2.074E+11	1.111E+10	8.83E-01	9.812E+09	9.71E-01	1.079E+10	1.12E+00	1.244E+10	1.28E+00	1.419E+10	9.67E-01	1.075E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
7.20	7.60	1.963E+11	1.031E+10	8.10E-01	8.356E+09	8.86E-01	9.137E+09	1.01E+00	1.046E+10	1.16E+00	1.199E+10	1.13E+00	1.168E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
7.60	8.00	1.859E+11	9.280E+09	7.49E-01	6.948E+09	8.15E-01	7.564E+09	9.27E-01	8.599E+09	1.06E+00	9.842E+09	1.15E+00	1.063E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
8.00	9.00	1.767E+11	2.011E+10	6.66E-01	1.340E+10	7.19E-01	1.446E+10	8.08E-01	1.625E+10	9.17E-01	1.844E+10	1.04E+00	2.086E+10	1.05E-01	2.105E+09	0.00E+00	0.000E+00	0.00E+00	0.00
9.00	10.00	1.565E+11	1.639E+10	5.84E-01	9.576E+09	6.20E-01	1.017E+10	6.81E-01	1.116E+10	7.60E-01	1.245E+10	8.75E-01	1.433E+10	8.01E-01	1.312E+10	0.00E+00	0.000E+00	0.00E+00	0.00
10.00	11.00	1.402E+11	1.413E+10	5.36E-01	7.576E+09	5.59E-01	7.901E+09	6.00E-01	8.471E+09	6.53E-01	9.227E+09	7.42E-01	1.048E+10	7.94E-01	1.122E+10	0.00E+00	0.000E+00	0.00E+00	0.00
11.00	12.00	1.260E+11	1.206E+10	5.05E-01	6.094E+09	5.20E-01	6.273E+09	5.48E-01	6.605E+09	5.83E-01	7.031E+09	6.44E-01	7.765E+09	7.00E-01	8.445E+09	6.14E-01	7.399E+09	0.00E+00	0.00
12.00	13.00	1.140E+11	1.034E+10	4.86E-01	5.023E+09	4.96E-01	5.124E+09	5.13E-01	5.306E+09	5.37E-01	5.553E+09	5.76E-01	5.959E+09	6.19E-01	6.398E+09	6.30E-01	6.513E+09	0.00E+00	0.00
13.00	14.00	1.036E+11	8.735E+09	4.72E-01	4.124E+09	4.79E-01	4.185E+09	4.91E-01	4.286E+09	5.06E-01	4.419E+09	5.31E-01	4.636E+09	5.57E-01	4.869E+09	5.77E-01	5.039E+09	0.00E+00	0.00
14.00	15.00	9.489E+10	7.776E+09	4.63E-01	3.599E+09	4.68E-01	3.639E+09	4.75E-01	3.695E+09	4.85E-01	3.768E+09	4.99E-01	3.883E+09	5.14E-01	3.997E+09	5.27E-01	4.098E+09	0.00E+00	0.00
15.00	16.00	8.712E+10	6.696E+09	4.57E-01	3.061E+09	4.60E-01	3.083E+09	4.65E-01	3.112E+09	4.70E-01	3.147E+09	4.78E-01	3.198E+09	4.84E-01	3.239E+09	4.88E-01	3.269E+09	0.00E+00	0.00
16.00	18.00	8.042E+10	1.137E+10	4.49E-01	5.105E+09	4.52E-01	5.133E+09	4.54E-01	5.165E+09	4.57E-01	5.193E+09	4.58E-01	5.209E+09	4.57E-01	5.191E+09	4.52E-01	5.141E+09	3.86E-01	4.39
18.00	20.00	6.906E+10	9.057E+09	4.39E-01	3.979E+09	4.41E-01	3.997E+09	4.43E-01	4.014E+09	4.44E-01	4.018E+09	4.41E-01	3.995E+09	4.34E-01	3.935E+09	4.24E-01	3.836E+09	3.83E-01	3.46
20.00	22.00	6.000E+10	7.441E+09	4.32E-01	3.214E+09	4.32E-01	3.217E+09	4.34E-01	3.227E+09	4.34E-01	3.227E+09	4.30E-01	3.197E+09	4.22E-01	3.141E+09	4.09E-01	3.045E+09	3.63E-01	2.70
22.00	24.00	5.256E+10	6.256E+09	4.25E-01	2.658E+09	4.24E-01	2.656E+09	4.25E-01	2.660E+09	4.25E-01	2.658E+09	4.21E-01	2.636E+09	4.13E-01	2.587E+09	4.01E-01	2.510E+09	3.55E-01	2.22
24.00	26.00	4.630E+10	5.193E+09	4.18E-01	2.170E+09	4.18E-01	2.169E+09	4.18E-01	2.169E+09	4.17E-01	2.166E+09	4.14E-01	2.149E+09	4.07E-01	2.112E+09	3.96E-01	2.054E+09	3.53E-01	1.83
26.00	28.00	4.111E+10	4.372E+09	4.11E-01	1.798E+09	4.11E-01	1.798E+09	4.11E-01	1.797E+09	4.10E-01	1.793E+09	4.07E-01	1.780E+09	4.01E-01	1.752E+09	3.91E-01	1.708E+09	3.53E-01	1.54
28.00	30.00	3.674E+10	3.814E+09	4.04E-01	1.542E+09	4.04E-01	1.542E+09	4.04E-01	1.541E+09	4.03E-01	1.537E+09	4.00E-01	1.527E+09	3.95E-01	1.507E+09	3.86E-01	1.474E+09	3.53E-01	1.34
30.00	34.00	3.292E+10	5.993E+09	3.92E-01	2.350E+09	3.92E-01	2.351E+09	3.92E-01	2.351E+09	3.92E-01	2.349E+09	3.90E-01	2.338E+09	3.86E-01	2.315E+09	3.80E-01	2.275E+09	3.53E-01	2.11
34.00	38.00	2.693E+10	4.589E+09	3.77E-01	1.730E+09	3.77E-01	1.731E+09	3.77E-01	1.731E+09	3.77E-01	1.731E+09	3.76E-01	1.725E+09	3.73E-01	1.714E+09	3.69E-01	1.695E+09	3.50E-01	1.60
38.00	42.00	2.234E+10	3.566E+09	3.63E-01	1.296E+09	3.63E-01	1.296E+09	3.64E-01	1.297E+09	3.64E-01	1.297E+09	3.63E-01	1.294E+09	3.61E-01	1.287E+09	3.58E-01	1.277E+09	3.44E-01	1.22
42.00	46.00	1.878E+10	2.822E+09	3.50E-01	9.865E+08	3.50E-01	9.866E+08	3.50E-01	9.876E+08	3.50E-01	9.886E+08	3.50E-01	9.889E+08	3.49E-01	9.837E+08	3.47E-01	9.787E+08	3.37E-01	9.51
46.00	50.00	1.595E+10	2.260E+09	3.36E-01	7.599E+08	3.36E-01	7.599E+08	3.36E-01	7.605E+08	3.37E-01	7.612E+08	3.37E-01	7.619E+08	3.36E-01	7.601E+08	3.35E-01	7.581E+08	3.29E-01	7.43
50.00	55.00	1.369E+10	2.255E+09	3.22E-01	7.265E+08	3.22E-01	7.265E+08	3.22E-01	7.269E+08	3.23E-01	7.277E+08	3.23E-01	7.288E+08	3.23E-01	7.290E+08	3.23E-01	7.284E+08	3.19E-01	7.19
55.00	60.00	1.144E+10	1.814E+09	3.08E-01	5.578E+08	3.08E-01	5.579E+08	3.08E-01	5.582E+08	3.08E-01	5.588E+08	3.09E-01	5.600E+08	3.09E-01	5.606E+08	3.09E-01	5.608E+08	3.07E-01	5.56
60.00	65.00	9.624E+09	1.432E+09	2.93E-01	4.200E+08	2.93E-01	4.202E+08	2.94E-01	4.206E+08	2.94E-01	4.211E+08	2.95E-01	4.223E+08	2.96E-01	4.233E+08	2.96E-01	4.239E+08	2.95E-01	4.23
65.00	70.00	8.192E+09	1.159E+09	2.79E-01	3.236E+08	2.79E-01	3.239E+08	2.80E-01	3.243E+08	2.80E-01	3.248E+08	2.81E-01	3.259E+08	2.82E-01	3.271E+08	2.83E-01	3.280E+08	2.84E-01	3.28
70.00	80.00	7.033E+09	1.736E+09	2.60E-01	4.520E+08	2.61E-01	4.524E+08	2.61E-01	4.530E+08	2.61E-01	4.538E+08	2.62E-01	4.554E+08	2.63E-01	4.572E+08	2.64E-01	4.589E+08	2.66E-01	4.61
80.00	90.00	5.298E+09	1.215E+09	2.37E-01	2.875E+08	2.37E-01	2.877E+08	2.37E-01	2.882E+08	2.38E-01	2.888E+08	2.39E-01	2.900E+08	2.40E-01	2.913E+08	2.41E-01	2.928E+08	2.43E-01	2.95
90.00	100.00	4.083E+09	8.637E+08	2.12E-01	1.835E+08	2.13E-01	1.837E+08	2.13E-01	1.841E+08	2.14E-01	1.846E+08	2.15E-01	1.856E+08	2.16E-01	1.866E+08	2.18E-01	1.879E+08	2.21E-01	1.90
100.00	130.00	3.219E+09	1.448E+09	1.75E-01	2.533E+08	1.75E-01	2.535E+08	1.75E-01	2.540E+08	1.76E-01	2.548E+08	1.77E-01	2.564E+08	1.78E-01	2.581E+08	1.80E-01	2.601E+08	1.83E-01	2.64
130.00	160.00	1.771E+09	6.992E+08	1.34E-01	9.365E+07	1.34E-01	9.368E+07	1.34E-01	9.386E+07	1.35E-01	9.410E+07	1.35E-01	9.455E+07	1.36E-01	9.515E+07	1.37E-01	9.578E+07	1.39E-01	9.75
160.00	200.00	1.071E+09	4.569E+08	1.05E-01	4.820E+07	1.06E-01	4.821E+07	1.06E-01	4.828E+07	1.06E-01	4.838E+07	1.06E-01	4.853E+07	1.07E-01	4.876E+07	1.07E-01	4.902E+07	1.09E-01	4.97
Equiv. 10 MeV Proton Fluence (cm ⁻²)					4.958E+12		9.167E+11		4.590E+11		2.682E+11		1.450E+11		8.405E+10		5.646E+10		2.75
P _{max} Equiv. 1 MeV Electron Fluence (cm ⁻²)					1.487E+16		2.750E+15		1.377E+15		8.046E+14		4.349E+14		2.522E+14		1.694E+14		8.25

Solar protons for I_{sc} (integration restricted to the input energy range). $D(E,...)$ values are coarse averages introduced into the table to simplify the output information; the accuracy of the results should not be affected by this simplification.

https://www.spenvis.oma.be/spenvis/htusers/k/kawakita/GEO_10Y/1778368789/spenvis_efp.html 7/8

2.40	2.60	4.946E+11	2.790E+10	1.03E+00	2.880E+10	7.81E-01	2.179E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.60	2.80	4.667E+11	2.458E+10	1.01E+00	2.481E+10	8.44E-01	2.074E+10	2.92E-03	7.177E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.80	3.00	4.421E+11	2.283E+10	9.80E-01	2.237E+10	8.79E-01	2.007E+10	8.96E-02	2.045E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.00	3.20	4.193E+11	2.028E+10	9.46E-01	1.919E+10	8.94E-01	1.813E+10	2.52E-01	5.116E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.20	3.40	3.990E+11	1.914E+10	9.11E-01	1.744E+10	8.92E-01	1.706E+10	4.09E-01	7.828E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.40	3.60	3.798E+11	1.739E+10	8.77E-01	1.525E+10	8.78E-01	1.528E+10	5.28E-01	9.184E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.60	3.80	3.624E+11	1.590E+10	8.44E-01	1.341E+10	8.59E-01	1.366E+10	6.11E-01	9.721E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.80	4.00	3.465E+11	1.444E+10	8.12E-01	1.172E+10	8.36E-01	1.207E+10	6.67E-01	9.624E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.00	4.20	3.321E+11	1.316E+10	7.84E-01	1.032E+10	8.12E-01	1.068E+10	6.99E-01	9.199E+09	1.48E-02	1.954E+08	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.20	4.40	3.189E+11	1.206E+10	7.60E-01	9.171E+09	7.86E-01	9.481E+09	7.15E-01	8.619E+09	1.26E-01	1.526E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.40	4.60	3.069E+11	1.121E+10	7.37E-01	8.260E+09	7.62E-01	8.537E+09	7.19E-01	8.055E+09	2.62E-01	2.942E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.60	4.80	2.957E+11	1.045E+10	7.14E-01	7.465E+09	7.39E-01	7.724E+09	7.15E-01	7.476E+09	3.71E-01	3.877E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.80	5.20	2.852E+11	1.916E+10	6.81E-01	1.306E+10	7.07E-01	1.355E+10	7.00E-01	1.342E+10	4.72E-01	9.039E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
5.20	5.60	2.661E+11	1.721E+10	6.41E-01	1.103E+10	6.67E-01	1.148E+10	6.75E-01	1.162E+10	5.50E-01	9.465E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
5.60	6.00	2.488E+11	1.548E+10	6.04E-01	9.354E+09	6.30E-01	9.753E+09	6.47E-01	1.001E+10	5.78E-01	8.951E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
6.00	6.40	2.334E+11	1.368E+10	5.73E-01	7.834E+09	5.95E-01	8.148E+09	6.17E-01	8.441E+09	5.82E-01	7.959E+09	3.20E-02	4.375E+08	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
6.40	6.80	2.197E+11	1.232E+10	5.46E-01	6.729E+09	5.66E-01	6.968E+09	5.87E-01	7.238E+09	5.73E-01	7.058E+09	2.38E-01	2.934E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
6.80	7.20	2.074E+11	1.111E+10	5.23E-01	5.813E+09	5.40E-01	6.001E+09	5.60E-01	6.225E+09	5.58E-01	6.196E+09	3.56E-01	3.950E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
7.20	7.60	1.963E+11	1.031E+10	5.04E-01	5.194E+09	5.19E-01	5.347E+09	5.37E-01	5.533E+09	5.40E-01	5.563E+09	4.13E-01	4.255E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
7.60	8.00	1.859E+11	9.280E+09	4.87E-01	4.516E+09	5.00E-01	4.638E+09	5.16E-01	4.785E+09	5.21E-01	4.834E+09	4.40E-01	4.083E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
8.00	9.00	1.767E+11	2.011E+10	4.63E-01	9.314E+09	4.74E-01	9.527E+09	4.86E-01	9.781E+09	4.92E-01	9.896E+09	4.48E-01	9.011E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
9.00	10.00	1.565E+11	1.639E+10	4.41E-01	7.223E+09	4.47E-01	7.322E+09	4.54E-01	7.433E+09	4.57E-01	7.498E+09	4.39E-01	7.188E+09	3.11E-01	5.099E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00
10.00	11.00	1.402E+11	1.413E+10	4.28E-01	6.055E+09	4.31E-01	6.086E+09	4.33E-01	6.118E+09	4.33E-01	6.113E+09	4.20E-01	5.930E+09	3.60E-01	5.093E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00
11.00	12.00	1.260E+11	1.206E+10	4.21E-01	5.082E+09	4.22E-01	5.085E+09	4.21E-01	5.074E+09	4.17E-01	5.027E+09	4.03E-01	4.859E+09	3.66E-01	4.417E+09	2.42E-01	2.921E+09	0.00E+00	0.000E+00
12.00	13.00	1.140E+11	1.034E+10	4.19E-01	4.333E+09	4.18E-01	4.319E+09	4.14E-01	4.279E+09	4.08E-01	4.214E+09	3.91E-01	4.048E+09	3.62E-01	3.742E+09	2.95E-01	3.048E+09	0.00E+00	0.000E+00
13.00	14.00	1.036E+11	8.735E+09	4.18E-01	3.654E+09	4.16E-01	3.637E+09	4.11E-01	3.593E+09	4.03E-01	3.520E+09	3.85E-01	3.363E+09	3.57E-01	3.119E+09	3.10E-01	2.704E+09	0.00E+00	0.000E+00
14.00	15.00	9.489E+10	7.776E+09	4.19E-01	3.256E+09	4.17E-01	3.239E+09	4.10E-01	3.192E+09	4.02E-01	3.122E+09	3.82E-01	2.971E+09	3.55E-01	2.757E+09	3.14E-01	2.445E+09	0.00E+00	0.000E+00
15.00	16.00	8.712E+10	6.696E+09	4.20E-01	2.815E+09	4.18E-01	2.797E+09	4.11E-01	2.753E+09	4.02E-01	2.690E+09	3.82E-01	2.556E+09	3.54E-01	2.372E+09	3.17E-01	2.125E+09	0.00E+00	0.000E+00
16.00	18.00	8.042E+10	1.137E+10	4.20E-01	4.777E+09	4.18E-01	4.752E+09	4.13E-01	4.690E+09	4.04E-01	4.591E+09	3.85E-01	4.371E+09	3.58E-01	4.064E+09	3.23E-01	3.668E+09	1.77E-01	2.01E+00
18.00	20.00	6.906E+10	9.057E+09	4.18E-01	3.788E+09	4.17E-01	3.776E+09	4.13E-01	3.740E+09	4.05E-01	3.672E+09	3.89E-01	3.520E+09	3.64E-01	3.298E+09	3.32E-01	3.009E+09	2.30E-01	2.08E+00
20.00	22.00	6.000E+10	7.441E+09	4.16E-01	3.094E+09	4.14E-01	3.079E+09	4.11E-01	3.058E+09	4.05E-01	3.014E+09	3.91E-01	2.907E+09	3.70E-01	2.757E+09	3.42E-01	2.549E+09	2.54E-01	1.89E+00
22.00	24.00	5.256E+10	6.256E+09	4.12E-01	2.577E+09	4.10E-01	2.565E+09	4.08E-01	2.550E+09	4.03E-01	2.520E+09	3.92E-01	2.451E+09	3.74E-01	2.342E+09	3.51E-01	2.195E+09	2.74E-01	1.71E+00
24.00	26.00	4.630E+10	5.193E+09	4.07E-01	2.114E+09	4.06E-01	2.109E+09	4.04E-01	2.097E+09	4.00E-01	2.076E+09	3.91E-01	2.030E+09	3.76E-01	1.954E+09	3.57E-01	1.853E+09	2.91E-01	1.51E+00
26.00	28.00	4.111E+10	4.372E+09	4.02E-01	1.759E+09	4.02E-01	1.756E+09	4.00E-01	1.748E+09	3.96E-01	1.732E+09	3.89E-01	1.699E+09	3.77E-01	1.647E+09	3.60E-01	1.575E+09	3.04E-01	1.33E+00
28.00	30.00	3.674E+10	3.814E+09	3.97E-01	1.513E+09	3.96E-01	1.511E+09	3.95E-01	1.506E+09	3.92E-01	1.494E+09	3.85E-01	1.470E+09	3.76E-01	1.433E+09	3.62E-01	1.380E+09	3.14E-01	1.19E+00
30.00	34.00	3.292E+10	5.993E+09	3.86E-01	2.313E+09	3.86E-01	2.311E+09	3.85E-01	2.306E+09	3.83E-01	2.297E+09	3.79E-01	2.270E+09	3.72E-01	2.227E+09	3.61E-01	2.165E+09	3.24E-01	1.94E+00
34.00	38.00	2.693E+10	4.589E+09	3.72E-01	1.709E+09	3.72E-01	1.708E+09	3.72E-01	1.706E+09	3.71E-01	1.702E+09	3.68E-01	1.688E+09	3.63E-01	1.666E+09	3.56E-01	1.636E+09	3.30E-01	1.51E+00
38.00	42.00	2.234E+10	3.566E+09	3.60E-01	1.283E+09	3.60E-01	1.283E+09	3.59E-01	1.282E+09	3.59E-01	1.280E+09	3.57E-01	1.272E+09	3.53E-01	1.260E+09	3.49E-01	1.244E+09	3.30E-01	1.17E+00
42.00	46.00	1.878E+10	2.822E+09	3.47E-01	9.782E+08	3.47E-01	9.782E+08	3.47E-01	9.783E+08	3.47E-01	9.780E+08	3.46E-01	9.759E+08	3.43E-01	9.676E+08	3.40E-01	9.589E+08	3.27E-01	9.22E+00
46.00	50.00	1.595E+10	2.260E+09	3.34E-01	7.541E+08	3.34E-01	7.541E+08	3.34E-01	7.542E+08	3.34E-01	7.543E+08	3.33E-01	7.537E+08	3.32E-01	7.502E+08	3.30E-01	7.460E+08	3.21E-01	7.25E+00
50.00	55.00	1.369E+10	2.255E+09	3.20E-01	7.215E+08	3.20E-01	7.216E+08	3.20E-01	7.218E+08	3.20E-01	7.222E+08	3.20E-01	7.223E+08	3.20E-01	7.214E+08	3.19E-01	7.191E+08	3.13E-01	7.05E+00
55.00	60.00	1.144E+10	1.814E+09	3.06E-01	5.545E+08	3.06E-01	5.546E+08	3.06E-01	5.549E+08	3.06E-01	5.553E+08	3.06E-01	5.558E+08	3.07E-01	5.559E+08	3.06E-01	5.552E+08	3.03E-01	5.48E+00
60.00	65.00	9.624E+09	1.432E+09	2.92E-01	4.177E+08	2.92E-01	4.179E+08	2.92E-01	4.183E+08	2.92E-01	4.188E+08	2.93E-01	4.196E+08	2.93E-01	4.203E+08	2.94E-01	4.204E+08	2.92E-01	4.18E+00
65.00	70.00	8.192E+09	1.159E+09	2.78E-01	3.219E+08	2.78E-01	3.222E+08	2.78E-01	3.226E+08	2.79E-01	3.232E+08	2.80E-01	3.241E+08	2.80E-01	3.250E+08	2.81E-01	3.257E+08	2.81E-01	3.25E+00
70.00	80.00	7.033E+09	1.736E+09	2.59E-01	4.502E+08	2.60E-01	4.505E+08	2.60E-01	4.512E+08	2.60E-01	4.520E+08	2.61E-01	4.535E+08	2.62E-01	4.551E+08	2.63E-01	4.565E+08	2.64E-01	4.58E+00
80.00	90.00	5.298E+09	1.215E+09	2.36E-01	2.868E+08	2.36E-01	2.871E+08	2.37E-01	2.875E+08	2.37E-01	2.881E+08	2.38E-01	2.893E+08	2.39E-01	2.906E+08	2.40E-01	2.91		